

4 Image Enhancement in the Frequency Domain

- Frequency domain processing techniques are based on modifying the Fourier Transform of an image
- The frequency domain refers to the plane of the two dimensional discrete Fourier transform of an image.
- The purpose of the Fourier transform is to represent a signal as a linear combination of sinusoidal signals of various frequencies
- Return to the original domain without losing any information via an inverse process.

Background

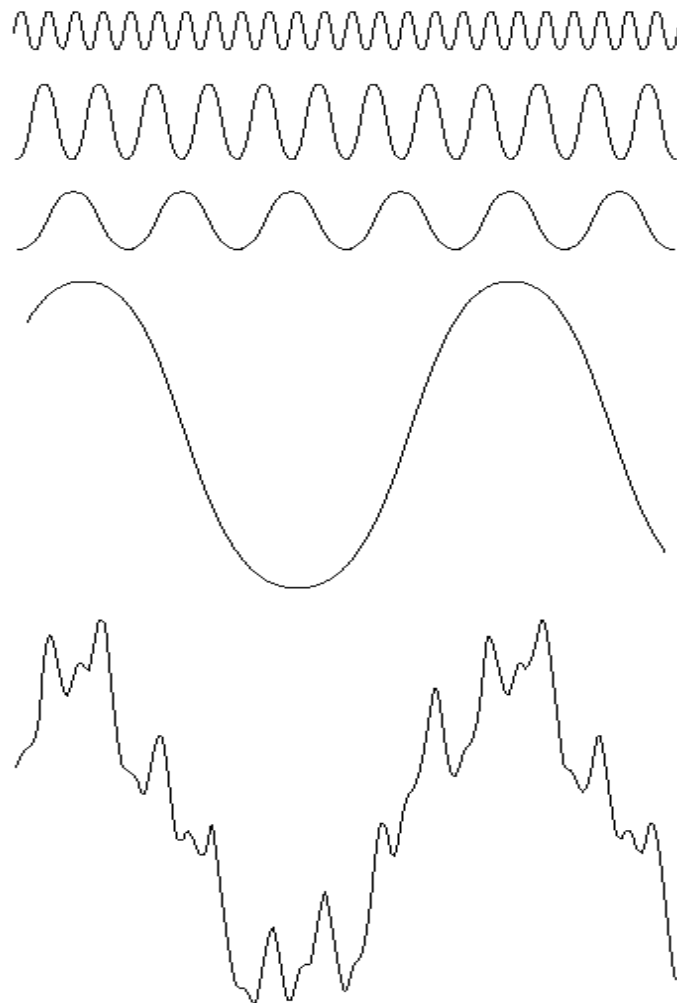


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Fourier Transform-Discrete Case

- The two-dimensional Fourier transform and its inverse
 - Fourier transform:

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$\text{for } u = 0, 1, 2, \dots, M-1, v = 0, 1, 2, \dots, N-1 \quad e^{j\theta} = \cos\theta + j\sin\theta$$

- Inverse Fourier transform:

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

$$\text{for } x = 0, 1, 2, \dots, M-1, y = 0, 1, 2, \dots, N-1$$

- u, v : the transform or frequency variables
 - x, y : the spatial or image variables

Basic Steps for Filtering

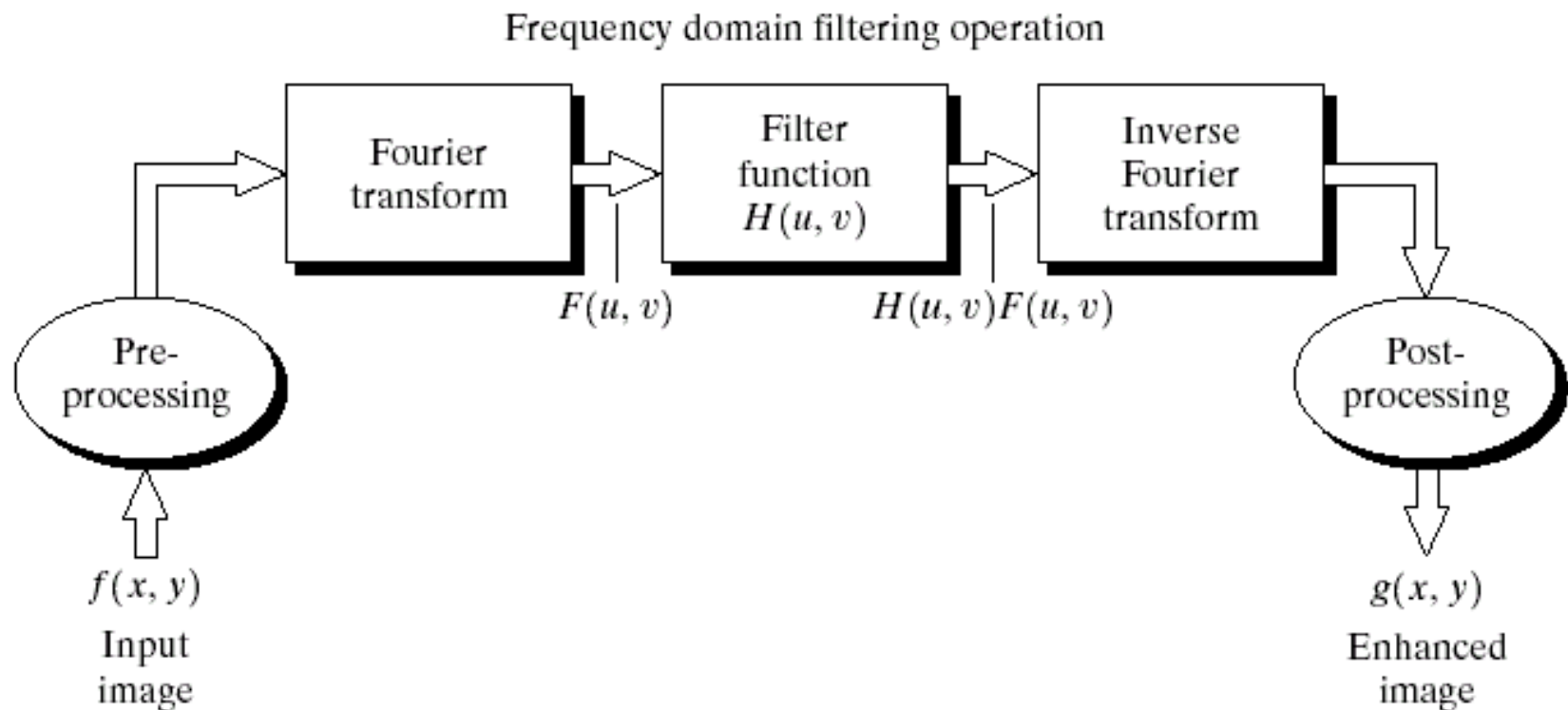


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Basic Model for Filtering

- The basic model for filtering in the frequency domain

$$G(u, v) = H(u, v)F(u, v)$$

where $F(u, v)$: the Fourier transform of the image to be smoothed

$H(u, v)$: a filter transfer function

Ideal Lowpass Filter (ILPF)

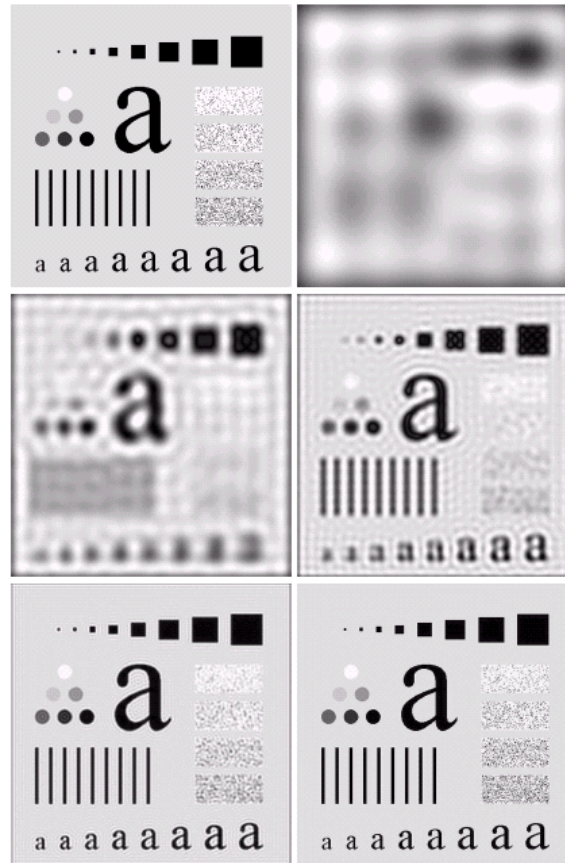
- The simplest lowpass filter is a filter that “cuts off” all high-frequency components of the Fourier transform that are at a distance greater than a specified distance D_0 from the origin of the transform.
- The transfer function of an ideal lowpass filter

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

where $D(u, v)$: the distance from point (u, v) to the center of the frequency rectangle

$$D(u, v) = \left[(u - M/2)^2 + (v - N/2)^2 \right]^{\frac{1}{2}}$$

Ideal Lowpass Filter Examples

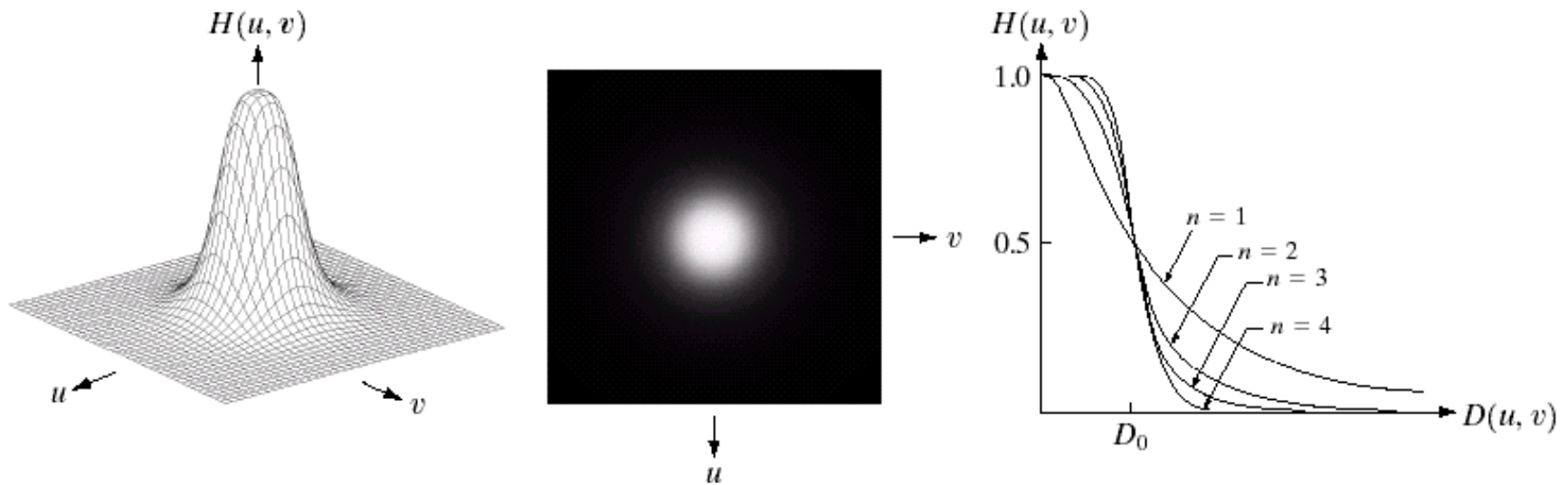


a	b
c	d
e	f

FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

Butterworth Lowpass Filter (BLPF)

$$H(u, v) = \frac{1}{1 + [D(u, v) / D_0]^{2n}}$$



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

Examples

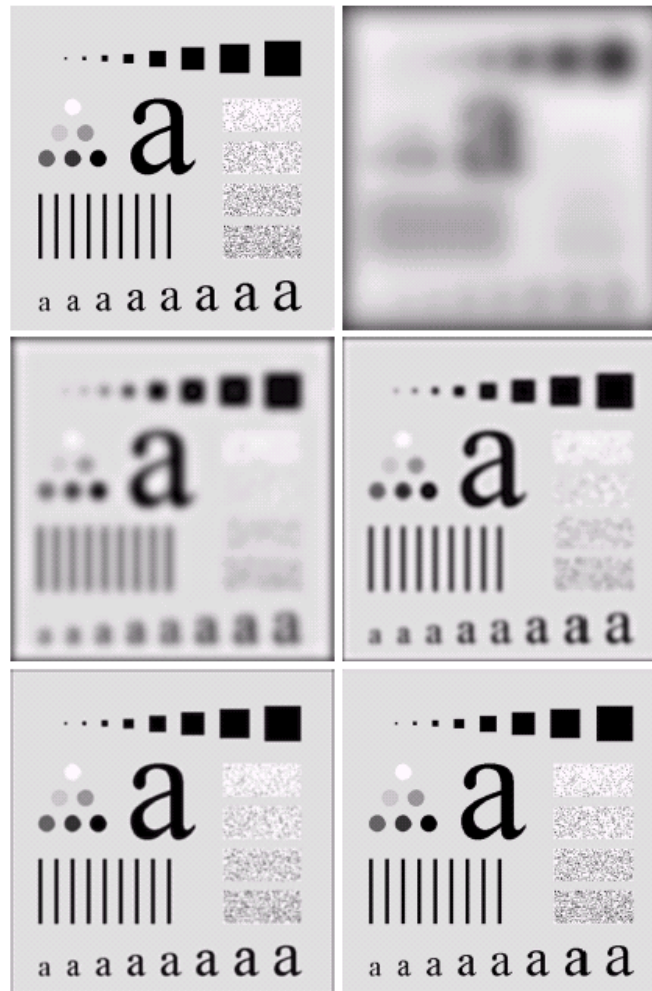
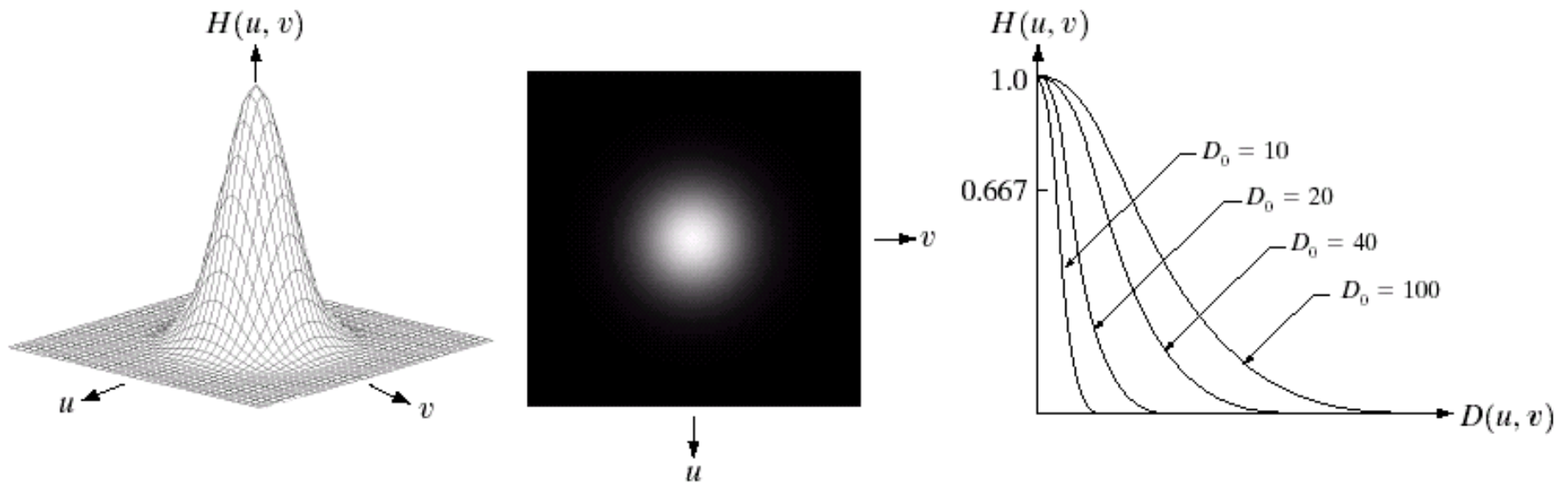


FIGURE 4.15 (a) Original image. (b)–(f) Results of filtering with BLPIs of order 2, with cutoff frequencies at radii of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Fig. 4.12.

Gaussian Lowpass Filter (GLPF)

$$H(u, v) = e^{-D^2(u, v) / 2D_0^2}$$



a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0 .

Gaussian Lowpass Filter Example

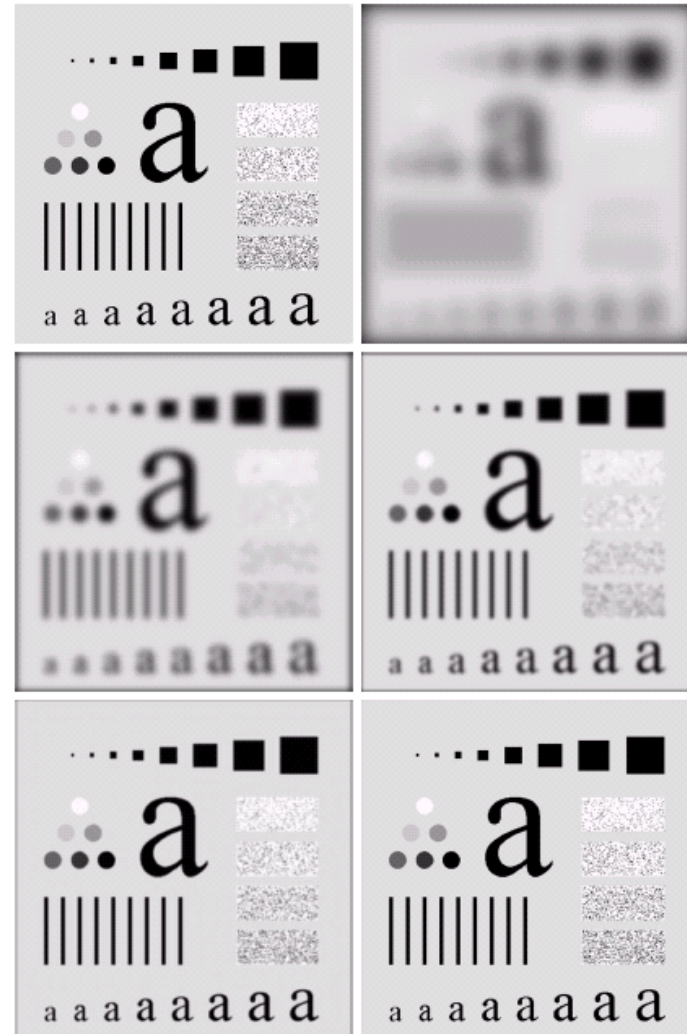


FIGURE 4.18 (a) Original image. (b)–(f) Results of filtering with Gaussian lowpass filters with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Figs. 4.12 and 4.15.

a b
c d
e f

Sharpening Frequency Domain Filter

$$H_{hp}(u, v) = H_{lp}(u, v)$$

Ideal highpass filter

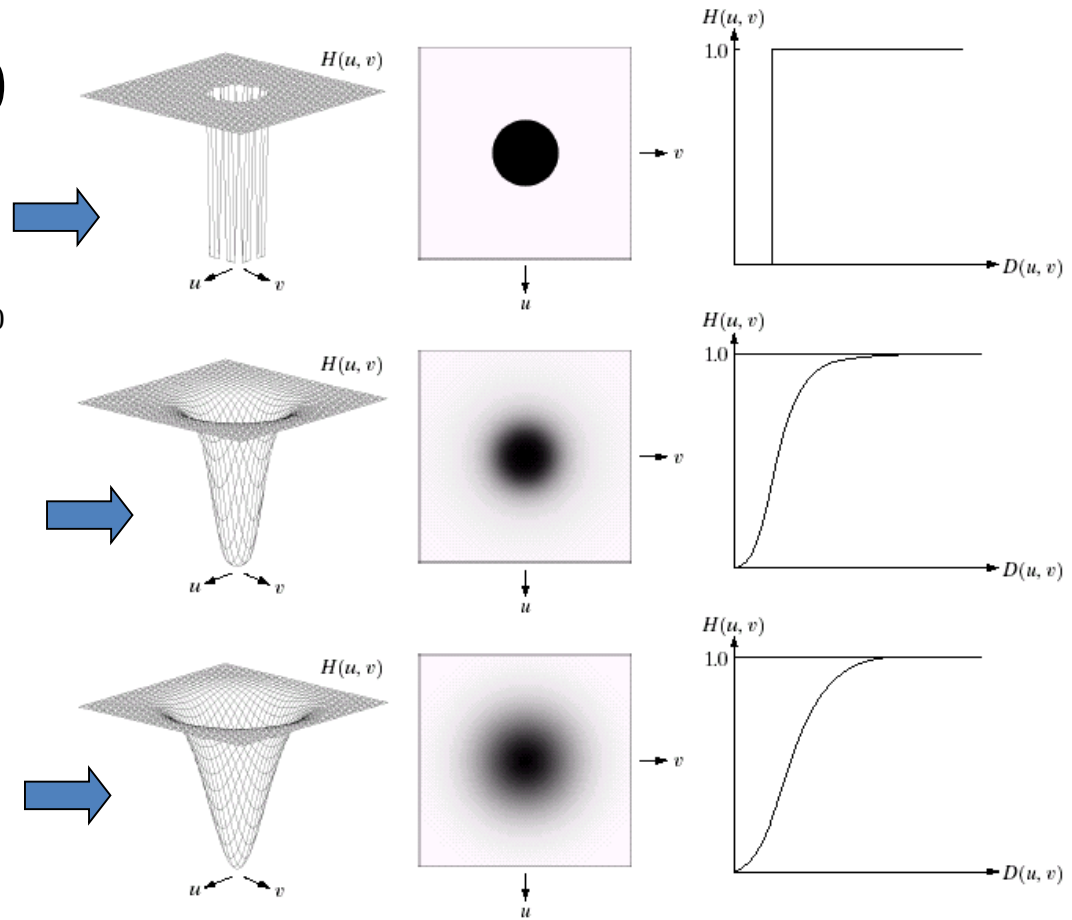
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

Butterworth highpass filter

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

Gaussian highpass filter

$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$

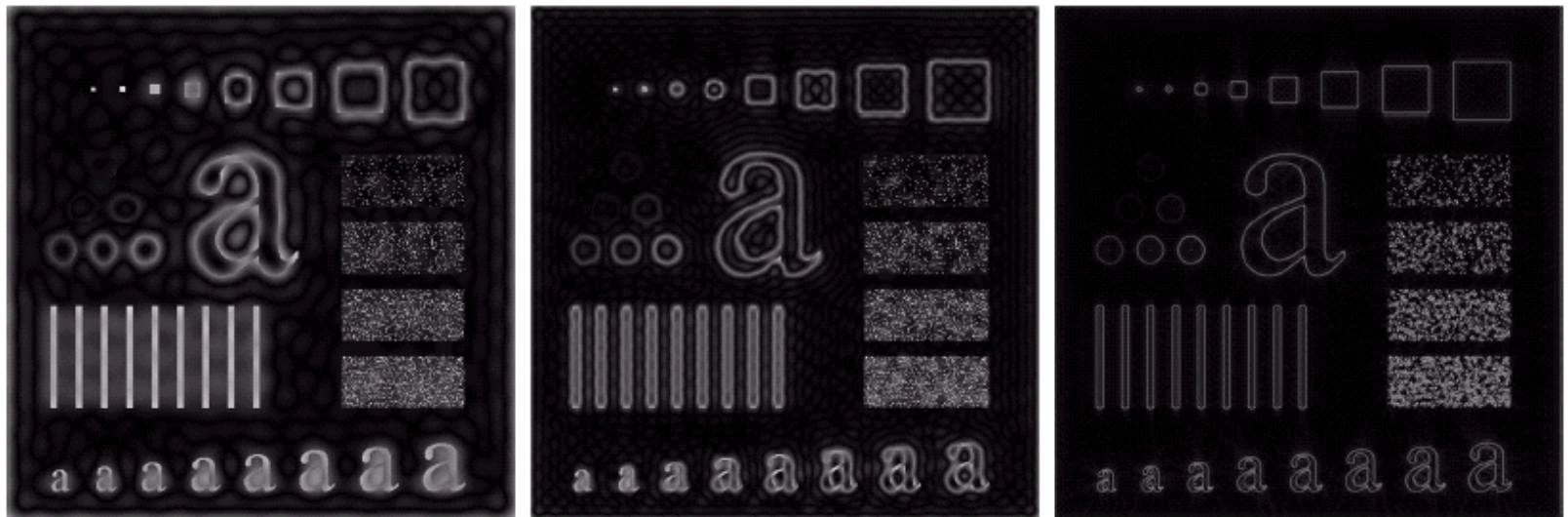


a b c
d e f
g h i

FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Ideal Highpass Filter (IHPF)

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$



a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15, 30$, and 80 , respectively. Problems with ringing are quite evident in (a) and (b).

Butterworth Highpass Filter (BHPF)

$$H(u, v) = \frac{1}{1 + [D_0 / D(u, v)]^{2n}}$$

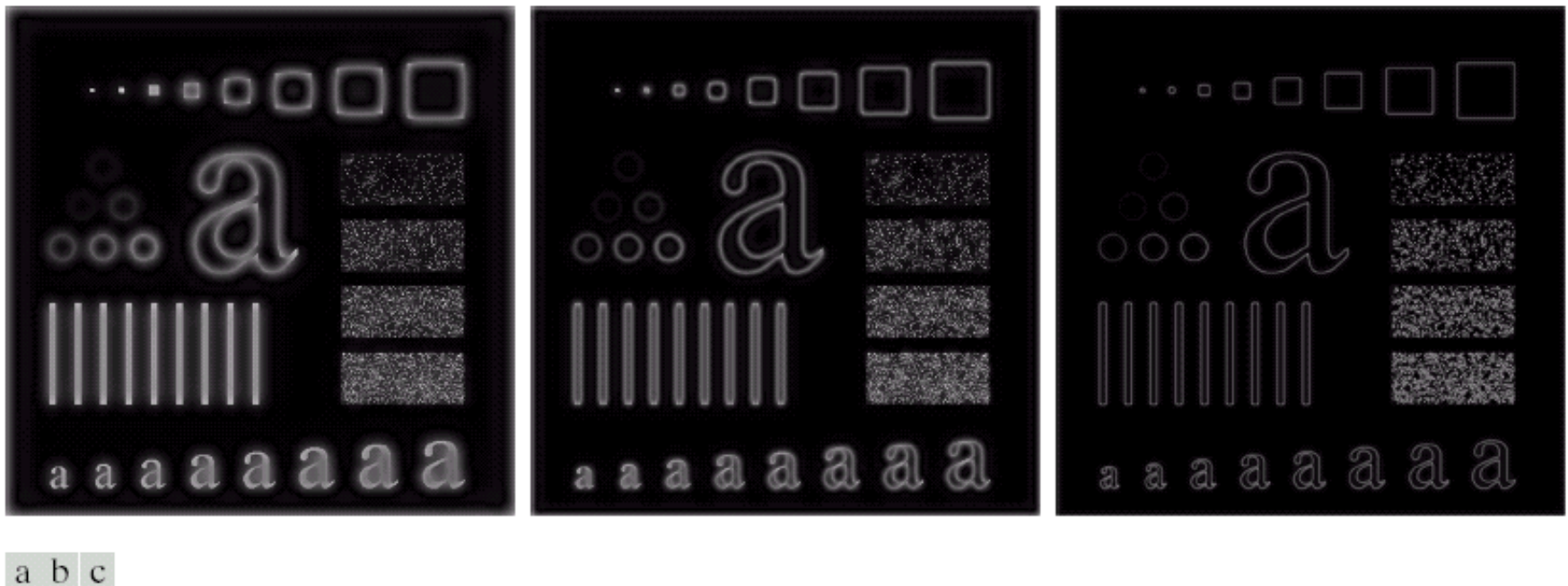


FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

Gaussian Highpass Filter (GHPF)

$$H(u, v) = 1 - e^{-D^2(u, v) / 2D_0^2}$$



a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.